Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

Name _____ Period _____ Date _____

Part I – Selected Response

1. Complete the conversion:
5 pounds = _____ ounces

A. 0.3
B. 21
C. 50
D. 80

2. Which numbers are equivalent to $4\frac{2}{5}$? Circle all that apply.

A. 4.25
B. 4.4
C. $\frac{22}{5}$
D. $4\frac{4}{10}$
E. $4.\overline{4}$
F. $\frac{5}{22}$

3. Nine of 21 players on a soccer team wore red jerseys. What is the ratio of players who wore red jerseys to players who did not wear red jerseys?

A. 3 : 7
B. 4 : 7
C. 4 : 3
D. 3 : 4

4. A bag holds only red and blue marbles. The ratio of red marbles to blue marbles is 4 : 1. What is the ratio of blue marbles to total marbles in the bag?

A. 1 : 4
B. 1 : 5
C. 4 : 5
D. 5 : 1

5. A train traveled 210 miles in 3.5 hours. What was its speed in miles per hour?

A. 60 miles per hour
B. 65 miles per hour
C. 70 miles per hour
D. 73.5 miles per hour

6. Which rate should be used to convert $\frac{3 \text{ gallons}}{1 \text{ day}}$ to $\frac{\text{gallons}}{\text{hour}}$?

A. $\frac{24 \text{ hours}}{1 \text{ day}}$
B. $\frac{1 \text{ day}}{24 \text{ hours}}$
C. $\frac{4 \text{ quarts}}{1 \text{ gallon}}$
D. $\frac{1 \text{ gallon}}{4 \text{ quarts}}$

7. For numbers 7a – 7d, circle YES or NO to tell whether the rate given is equivalent to a pump moving 900 gallons per hour.

7a. 15 gallons per minute	**YES**	**NO**
7b. 54,000 gallons per minute	**YES**	**NO**
7c. 21,600 gallons per day	**YES**	**NO**
7d. 0.25 gallon per second	**YES**	**NO**

8. What is the unit rate for $\frac{\frac{4}{5} \text{ liter}}{2 \text{ minutes}}$?

A. $1\frac{3}{5}$ liters per minute
B. $2\frac{1}{2}$ liters per minute
C. $\frac{4}{5}$ liter per minute
D. $\frac{2}{5}$ liter per minute

9. Jada hiked $4\frac{2}{3}$ miles in $1\frac{3}{4}$ hours. Assume she hiked the entire time at the same rate. What was her speed in miles per hour?

A. $8\frac{1}{6}$ miles per hour
B. $\frac{3}{8}$ mile per hour
C. $2\frac{2}{3}$ miles per hour
D. $3\frac{1}{2}$ miles per hour

Form C

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Part II – Constructed Response

Complete each conversion.

1. 7 yards = _____ feet

2. 9 centimeters = _____ millimeters

3. 3,000 pounds = _____ tons

4. 54 inches = _____ feet

Convert each fraction to a decimal.

5. $\frac{9}{20}$

6. $\frac{5}{6}$

Convert each decimal to a fraction.

7. 0.8

8. 0.24

9. Simplify the ratio 25 : 15.

10. Ten of 25 runners on the cross-country team finished the race in under 30 minutes. Write three different simplified number ratios using this information. Explain the meaning of each ratio using words.

Ratio (numbers): Explanation:	Ratio (numbers): Explanation:	Ratio (numbers): Explanation:

Find the unit rate.

11. $\frac{72 \text{ feet}}{6 \text{ seconds}}$

12. $\frac{\frac{3}{5} \text{ gallon}}{\frac{9}{10} \text{ minute}}$

13. Grace spent $8.75 on 5 notebooks. Omar spent $13.20 on 8 notebooks. Which person paid less per notebook? Show all work necessary to justify your answer.

14. A deer runs at a rate of 30 miles per hour. Write this rate as feet per hour. Show all work necessary to justify your answer.

15. A cyclist rides at a rate of 21 kilometers per hour. Write this rate as meters per minute. Show all work necessary to justify your answer.

16. Nina walked $\frac{2}{3}$ mile in $\frac{1}{4}$ hour. Assume she walked the entire time at the same rate. What was her speed in miles per hour? Caleb answered this question, but made a mistake. Explain Caleb's mistake. Then complete the work correctly to answer the question.

I need to find: $\frac{\frac{2}{3} \text{ mile}}{\frac{1}{4} \text{ hour}}$

$\frac{\frac{2}{3} \text{ mile}}{\frac{1}{4} \text{ hour}} = \frac{2}{3} \div \frac{1}{4}$

$= \frac{3}{2} \cdot \frac{1}{4}$

$= \frac{3}{8}$ mile per hour

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Part III – Problem Solving

The science club has 8 seventh graders and 18 eighth graders. They fold paper cranes for a hospital fundraiser, and a sponsor donates money for each crane. Each seventh grader earned $3.00 per crane and folded at a rate of 15 cranes per hour. Each eighth grader earned $2.00 per crane and folded at a rate of 4 cranes every 20 minutes. What is the ratio of the amount of money the seventh graders raised in one hour to the amount of money the eighth graders raised in one hour?